KPlaceNet: last-minute cheat sheet

Say this first

We estimate a photograph's geographic region from pixels. ResNet-50 produces 2,048 features, a classifier and optional training-only KNN vote choose one of 300 cells, and we display its centroid. Kerala-area mode assumes that domain; it cannot reject foreign scenes and its bounding rectangle includes neighboring territory.

Procedure

RGB image; deterministic resize and 224 crop; ImageNet normalization; ResNet-50; 300 logits; softmax; optional 80/20 blend with five cosine-nearest training features; highest-scoring cell; saved centroid; map. Training GPS creates labels. Inference uses pixels. Gallery click loads an image; Locate photo runs inference.

Final measured result

357 official test photos. L5 / L6 / final hybrid: within 25 km 41.18% / 47.62% / 53.50%; median error 42.08 / 28.41 / 21.84 km. Hybrid mean 80.65 km; p90 worsens versus L6. L7 reuses the L6 benchmark, so it is a follow-up, not fresh independent confirmation.

What L6 changed

Freeze the L5 backbone and cells. Train only the 614,700-parameter head. Compare uniform and geographic smoothing. q = 0.9 one-hot + 0.1 softmax(-centroid distance / 15 km). AdamW, learning rate 0.0001, weight decay 0.01, 12 epochs. Geographic recipe wins validation at epoch 12 before test evaluation.

The split problem

15,939 train; 1,000 random validation; 357 official test. IDs disjoint. In our recovered subset, 982 validation photos are within 1 km of training, versus zero official test photos. Random validation is optimistic.

Do not confuse

K-means builds coordinate groups; KNN retrieves five training-only ResNet feature neighbors. Raw confidence is not measured correctness. Coverage is not top-1 accuracy. Bounds are not a confidence interval. Four decimal places are not GPS precision. Google Maps displays the estimate and does not correct it.

Uncertainty and contribution

Kerala score is uncalibrated. Global legacy temperature scaling has unverified artifact identity. Separate 20k global study: ECE 0.4171 to 0.0453, coverage 90.8%, average set 159.7/300. Conformal rank: ceil((n+1)(1-alpha)); rank beyond n means infinity. My contributions include evaluation integrity, reproducibility, integration and L6/L7 follow-ups; credit the existing project separately.

Demo and ETH comparison

Show Kottayam example, Locate photo, coordinates, raw score, Google Maps, then the Southern Kerala failure. ETH paper: prototypes plus aerial embeddings and geographic interpolation. Our L6 adapts boundary awareness only. The 350 Commons records are future-data candidates, not added training images.